

Education

Master of Science in Applied Physics, City University of Hong Kong

Hong Kong, China

Dissertation Supervisor: Xiangqiang CHU

Sep. 2023 – Oct. 2024

Bachelor of Science in Physics, Lanzhou University (One of China's key national universities)

Lanzhou, China

Specialization: Metallic Materials Physics; Undergraduate Thesis Supervisor: Dongsheng LEI Sep. 2019 – Jul. 2023

Selected Honor: Outstanding Student League Cadre (2020 & 2021; Top 5%)

Publications

- [1] Shuowen Li, Danqi Huang, **Fangyu Hou**, Qiuyu Lu, Haipeng Mi, and Zhihao Yao. "Computational Design of 3D-Printable Freeform Caustic Surfaces." *ACM SIGGRAPH 2026 Posters*, Article 4 (2026).
[Link](#). DOI: [10.1145/3799825.3818720](#).
- [2] Yanguang Zhu and **Fangyu Hou**. "Cotton Smoldering Simulation and Multimodal Gas Detection Device." Chinese invention patent application, pending.

Research Experience

Multimodal Olfactory Sensing and Computing

Jan. 2026 – Present

Research Assistant, The Future Lab, Tsinghua University

Beijing, China

- Led a cotton-smoldering early-warning study combining field investigations at two storage facilities, a five-factor AHP-based risk model, and a seven-temperature (65–215 °C) TG-FTIR-GC-MS characterization protocol to guide multimodal sensor and experimental-system design; developed the system concept underlying a Chinese invention patent application filed in July 2026.
- Optimized thin-film sensing layers and electrode/substrate architectures using spin coating, layer-by-layer assembly, and ultrasonic spray coating; conducted room-temperature gas-sensing tests with an in-house four-channel electronic nose, improving NH₃ sensing from poorly recoverable single responses to reproducible sensing cycles at 10–25 ppm, while achieving measurable responses to H₂S and CH₃SH.
- Co-developed E-Nose Station, an electronic-nose experimental analysis platform, developing data-acquisition interfaces for response-curve visualization, cross-experiment comparison, odor-fingerprint analysis, and model-assisted data screening; supported semi-automated experiments through software-based acquisition control, timing, and stage reminders, and applied computational olfaction methods to complex gas-sensing tasks including abnormal oral-gas identification, insect pheromone detection, and indoor pollutant-emission monitoring.

Biomimetic Colored Radiative-Cooling Films

Sep. 2025 – Dec. 2025

Research Assistant, Great Bay University

Dongguan, China

- Fabricated and characterized microstructured colored radiative-cooling films using vacuum deposition, UV-Vis spectroscopy, infrared/thermal imaging, nanoindentation, and microscopy; processed spectral and imaging data to compare material performance and analyze structure–property relationships among microstructure, visible appearance, optical transparency, and thermal behavior.

GSDM Protein Dynamics and Cell Pyroptosis

Sep. 2024 – Dec. 2024

Research Assistant, Institute of Physics, Chinese Academy of Sciences

Beijing, China

- Investigated GSDM-family protein-mediated pyroptosis using HS-AFM to visualize pore-forming dynamics, performing experiments with ThGSDM, GSDMD, and GSDME on bare mica and DOPC/DOPS supported lipid bilayers; also contributed to X-ray crystallography data processing.

Flexible-Substrate Metasurface Optical Tweezers

Mar. 2022 – Mar. 2023

National Undergraduate Innovation and Entrepreneurship Training Program

Lanzhou, China

- Designed flexible-substrate metasurface lenses using PDMS and nanopillar unit cells, combining FDTD simulations with MATLAB-based optical-force modeling to characterize phase response, focal fields, and three-dimensional particle trapping.

- Modeled metasurface deformation under varying substrate stretching conditions and demonstrated stable simulated trapping of 15 nm polystyrene spheres near the self-focusing region.

Human-Centered AI Projects

Locus: Human–AI Decision Support for PhD Research Fit and Advisor Discovery
Independent Project

Apr. 2026 – Present
Beijing, China

- * Project results at [link 1](#).
- Independently designed and developed Locus, a Human–AI decision-support platform that models users’ academic backgrounds, research evidence, interests, and constraints for personalized PhD research positioning and advisor exploration.
- Built an adaptive advisor-search and ranking engine combining offline faculty retrieval, LLM-generated candidates, research-similarity scoring, and institution/discipline constraints, with dynamic search expansion under varying candidate yields.
- Designed uncertainty-aware verification mechanisms using Web, OpenAlex, and ORCID evidence, and piloted Locus with 50+ users to iteratively refine its recommendation and interaction workflows.

Skills

Programming & Data Analysis:	Python (NumPy, Pandas, Matplotlib, SciPy), MATLAB, SQLite
Web Development & Deployment:	JavaScript, HTML/CSS, Flask, Linux, Git, Nginx, Web Deployment, DNS, SSL/TLS Configuration
Scientific Computing & Engineering:	FDTD Solutions, COMSOL, SolidWorks, Arduino
Sensing & Experimental Methods:	HS-AFM/AFM, Spin Coating, Ultrasonic Spray Coating, Layer-by-Layer Assembly, Materials Characterization (Optical, Thermal, Mechanical)